Two sample z/t test

Dataset: Setup

- We have used synthetic data for now, so we know the true shift.
- Reference sample $X \sim \mathcal{N}(0, I)$: each value is drawn from a bell curve with mean 0 and std 1.
- Each data point has $d = 10$ **features** (columns).
- Each sample has $n \in \{10, 20, 50, 100\}$ data points (rows).
- A new sample Y is drawn with one thing changed.
- Each setting is repeated 100 times.

	feature 1	feature 2	$\cdots$	feature 10
point 1	0.3	-1.2	$\cdots$	0.0
$\vdots$	$\vdots$	$\vdots$		$\vdots$
point n	0.5	-0.3	$\cdots$	-0.8

Dataset: Shifts

Shift	**What changes in Y**
Mean	Each feature's mean shifts by $a/\sqrt{10}$.
Variance	Each feature's standard deviation is multiplied by s.
Heavy-tailed mean	Same mean shift, with Student-t_3 noise.

- a **(mean shift):** Controls the shift magnitude. Levels: $a \in \{0, 0.4, 0.8, 1.2, 1.6\}$.
- s **(scale shift):** Controls the spread. For example, $s = 1.2$ increases the standard deviation by 20%.
- **No shift:** $a = 0$ and $s = 1$; used to measure false alarms.
- **Heavy-tailed noise:** Produces more extreme values and tests robustness to violations of the noise assumptions.

Tests ($\alpha = 0.05$)

- **t-test**: compares the mean of each feature.
- **KS test**: compares the full distribution of eaution assumption).
- t-test and KS: report a shift if **any** of the 10 features has $p < 0.05$.
- **GLRT**: likelihood ratio under a Gaussian model.
- GLRT: reports a shift if $-2 \log \Lambda$ is above the 95th percentile of χ^2_{20}.

Introduction

- The two-sample z/t-test examines whether two populations have the same mean, using their observed samples.
- H_0: $\mu_x = \mu_y$
- H_1 : $\mu_x \neq \mu_y$
- Two different distributions can have the same population mean.
  1. $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 4)$
  2. $X \sim \mathcal{N}(2, 1)$ and $Y \sim \text{Exp}(\lambda = 1/2)$
- But, in some cases we might have some prior knowledge of the samples and know that both belong to the same distribution family (ex: Gaussian), and have the same population variance.
  - In this case, if $\mu_x = \mu_y$, both distributions are same - what we want to check.

- Even if two populations have the same mean, two randomly drawn samples will generally have different sample means due to sampling variability.
- We want to determine whether the observed difference between the sample means is statistically significant enough to reject H_0.

When is a mean difference significant?

- First we measure how much the sample means differ: $\bar{X} - \bar{Y}$
- Let's say we get D $= 5$.
  1. *Situation A*: Values usually between 49 & 51, so D $= 5$ is relatively large.
  2. *Situation B*: Values usually between 0 & 100, so D $= 5$ may be relatively small.
- The observed difference alone is insufficient. We must account for the natural variability of the sample means.
- We quantify this variability using the **standard error.**

Standard Error

- **Standard deviation** measures how spread out individual observations are around their mean.
- However, we are comparing sample means, not individual observations.
- Even if the population means are equal, the sample means can differ due to random sampling.
- Standard error (SE) measures how much a sample statistic (such as the sample mean) would vary across repeated random samples.

$$SE(\bar{X} - \bar{Y}) = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

Test Statistic

$$t = \frac{\bar{X} - \bar{Y}}{SE(\bar{X} - \bar{Y})}$$

- The further away this value is from 0, the more likely it is that we can reject H_0
- But how do we determine whether the observed t-value is sufficiently extreme to reject H_0? We calculate its **p-value.**
- Assuming H_0 is true, the p-value is the probability of observing a test statistic at least as extreme as the one obtained.
- We compare the p-value against a chosen significance level, α (typically 0.05).

$$\begin{aligned} p < \alpha &\implies \text{Reject } H_0 \\ p \geq \alpha &\implies \text{Fail to reject } H_0 \end{aligned}$$

Z-test

- The difference between z-test and t-test is that in z-test, the population variances are known.

$$z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}}$$

KS Test: Introduction

- The Kolmogorov–Smirnov (KS) test examines whether two populations have the same **distribution**, not just the same mean, using their observed samples.
- H_0**:** $F_X = F_Y$
- $H_1 : F_X \neq F_Y$
- Here F is the cumulative distribution function (CDF): $F(x)$ is the probability that a value is at most x.
- The t-test cannot separate $X \sim \mathcal{N}(0, 1)$ from $Y \sim \mathcal{N}(0, 4)$, since both have mean 0. The KS test can, because their distributions differ.
- The KS test is **nonparametric**: it does not assume that the data are Gaussian or that the variances are equal.

Empirical CDF

- We do not know the true CDF, so we estimate it from a sample by counting.
- The **empirical CDF** is the fraction of the sample that is at most x:

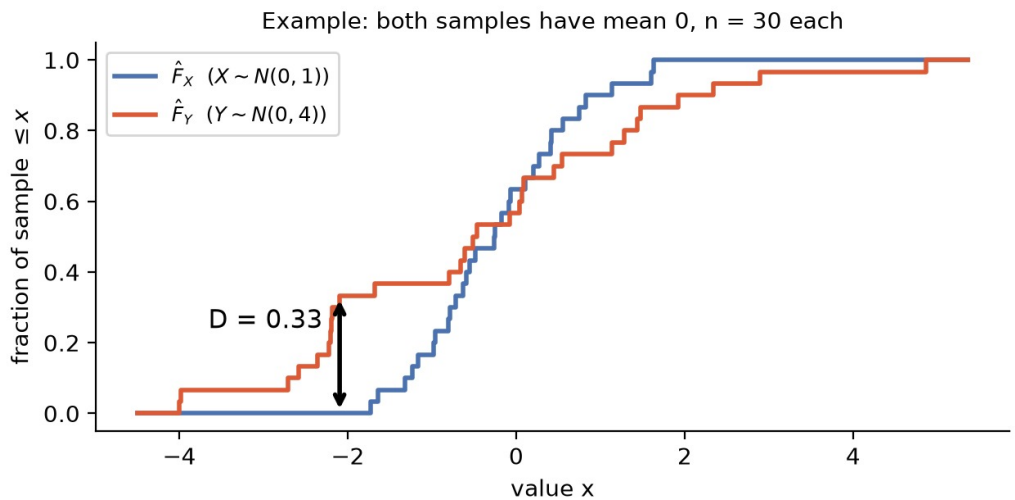
$$\hat{F}_X(x) = \frac{1}{n_X} \sum_{i=1}^{n_X} \mathbf{1}\{X_i \leq x\}$$

- Here $\mathbf{1}\{\cdot\}$ is 1 when the condition is true and 0 otherwise.
- Example: for $X = \{1, 2, 4, 7\}$, $\hat{F}_X(3) = \frac{2}{4} = 0.5$.
- It is a staircase: it starts at 0, rises by $1/n_X$ at each data point, and ends at 1.
- Even if H_0 is true, $\hat{F}_X$ and $\hat{F}_Y$ will differ slightly due to sampling variability.

KS Statistic

$$D = \max_x \left| \hat{F}_X(x) - \hat{F}_Y(x) \right|$$

- D is the largest vertical gap between the two empirical CDFs, so $0 \leq D \leq 1$.
- The further D is from 0, the more likely it is that we can reject H_0.
- Both CDFs only change at data points, so it is enough to check the gap at the data points.



KS Test: p-value and Decision

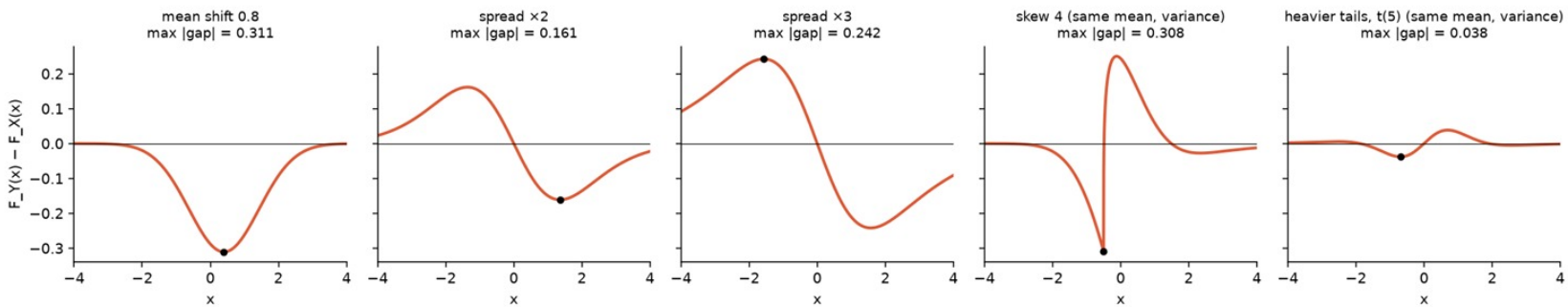
- How large must D be? We compare it with how large it would typically be if H_0 were true, just as the t-test compares the difference of means with the standard error.
- For continuous data, the distribution of D under H_0 is the same whatever the data distribution is, so no Gaussian assumption is needed.
- For large samples, we reject H_0 at $\alpha = 0.05$ if

$$\boxed{\sqrt{\frac{n_X n_Y}{n_X + n_Y}}\, D > 1.358}$$

- The factor $\sqrt{n_X n_Y/(n_X + n_Y)}$ rescales D for the sample sizes, similar to dividing by the standard error in the t-test.
- For small samples the exact distribution of D is used. The p-value is the probability, assuming H_0 is true, of observing a D at least as large as the one obtained, and the decision rule is the same: reject H_0 if $p < \alpha$.

What the KS Test Can See

- The KS test reacts to **any** change in the distribution: a shift in the mean, a change in the variance, or a change in shape.
- The t-test reacts only to a change in the mean. If the mean is unchanged, it cannot see a variance shift.
- The price of this generality: for a pure mean shift in Gaussian data, the t-test is the more powerful test, since it uses exactly the information that matters.
- In our experiment, the KS test is applied to each of the 10 features separately, and we report a shift if **any** feature has $p < 0.05$ (the same rule as the t-test).



GLRT: Introduction

- The generalised likelihood ratio test (GLRT) asks: how much better do the data fit if each sample gets its **own** Gaussian, instead of one shared Gaussian?
- H_0: $(\mu_X, \Sigma_X) = (\mu_Y, \Sigma_Y)$ (one Gaussian for both samples)
- H_1: $(\mu_X, \Sigma_X) \neq (\mu_Y, \Sigma_Y)$ (a separate Gaussian for each)
- The likelihood ratio compares the best fit under each hypothesis:

$$\Lambda = \frac{\max_{H_0} L}{\max_{H_1} L} \leq 1, \qquad \boxed{G = -2 \log \Lambda}$$

- Λ near 1 (G near 0): one Gaussian fits as well as two, so no evidence of a shift. Large G: reject H_0.
- Unlike the t-test, it sees mean **and** covariance changes.

Step 1: Write the Gaussian likelihood

For one observation $x \in \mathbb{R}^d$:

$$p(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right]$$

- $(2\pi)^{-d/2}$ is a dimensional normalisation factor.
- $|\Sigma|^{-1/2}$ accounts for the volume and spread of the Gaussian.
- The exponential penalises observations far from the mean in covariance-adjusted distance.

For independent observations, likelihoods multiply, so log-likelihoods add. For N observations $z_1, \ldots, z_N$:

$$\ell(\mu, \Sigma) = -\frac{Nd}{2}\log(2\pi) - \frac{N}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{N}(z_i-\mu)^\top \Sigma^{-1}(z_i-\mu)$$

This is our starting point.

Step 2: Maximum-likelihood estimates

$$\boxed{\hat{\mu} = \bar{z} = \frac{1}{N}\sum_{i=1}^{N} z_i} \qquad \boxed{\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^{N}(z_i - \bar{z})(z_i - \bar{z})^\top}$$

- The denominator is N, not $N-1$: this is the Gaussian MLE, not the unbiased sample covariance.

Substituting the fitted parameters, the quadratic terms collapse (using $a^\top B a = \mathrm{tr}(B a a^\top)$):

$$\sum_{i=1}^{N}(z_i-\bar{z})^\top \hat{\Sigma}^{-1}(z_i-\bar{z}) = \mathrm{tr}\left[\hat{\Sigma}^{-1}\textstyle\sum_i (z_i-\bar{z})(z_i-\bar{z})^\top\right] = \mathrm{tr}\left[\hat{\Sigma}^{-1} N\hat{\Sigma}\right] = Nd$$

So the maximised log-likelihood is

$$\boxed{\hat{\ell} = -\frac{N}{2}\left[d\log(2\pi) + d + \log|\hat{\Sigma}|\right]}$$

After fitting, the likelihood depends on the data **only through** $\log|\hat{\Sigma}|$.

Steps 3–4: Likelihood under H_0 and H_1

Under H_0, pool all $N = n + m$ observations and fit one Gaussian:

$$\bar{z} = \frac{1}{N}\sum_{i=1}^{N} z_i, \qquad \hat{\Sigma}_0 = \frac{1}{N}\sum_{i=1}^{N}(z_i - \bar{z})(z_i - \bar{z})^\top$$

$$\hat{\ell}_0 = -\frac{N}{2}\Big[d\log(2\pi) + d + \log|\hat{\Sigma}_0|\Big]$$

- $\hat{\Sigma}_0$ is taken about the grand mean, so a mean difference between X and Y shows up as **extra spread**.

Under H_1, fit X and Y separately; their log-likelihoods add:

$$\hat{\ell}_1 = -\frac{n}{2}\Big[d\log(2\pi) + d + \log|\hat{\Sigma}_X|\Big] - \frac{m}{2}\Big[d\log(2\pi) + d + \log|\hat{\Sigma}_Y|\Big]$$

Step 5: The test statistic

Since $\log \Lambda = \hat{\ell}_0 - \hat{\ell}_1$, we obtain $G = 2(\hat{\ell}_1 - \hat{\ell}_0)$.

Substituting the fitted log-likelihoods, all constant terms cancel ($N = n + m$), leaving

$$\boxed{G = N \log |\hat{\Sigma}_0| - n \log |\hat{\Sigma}_X| - m \log |\hat{\Sigma}_Y|}$$

- This is the Gaussian likelihood-ratio statistic for testing equality of two multivariate Gaussians.
- Intuition: if X and Y come from the same Gaussian, pooling costs nothing and $|\hat{\Sigma}_0|$ is close to the separate determinants. A shift in mean or covariance inflates $|\hat{\Sigma}_0|$, so G grows.
- **Diagonal case** (features assumed independent, as in our experiment): $|\Sigma| = \prod_j \sigma_j^2$, so

$$G = \sum_{j=1}^{d} \left[N \log \hat{\sigma}_{0j}^2 - n \log \hat{\sigma}_{Xj}^2 - m \log \hat{\sigma}_{Yj}^2 \right]$$

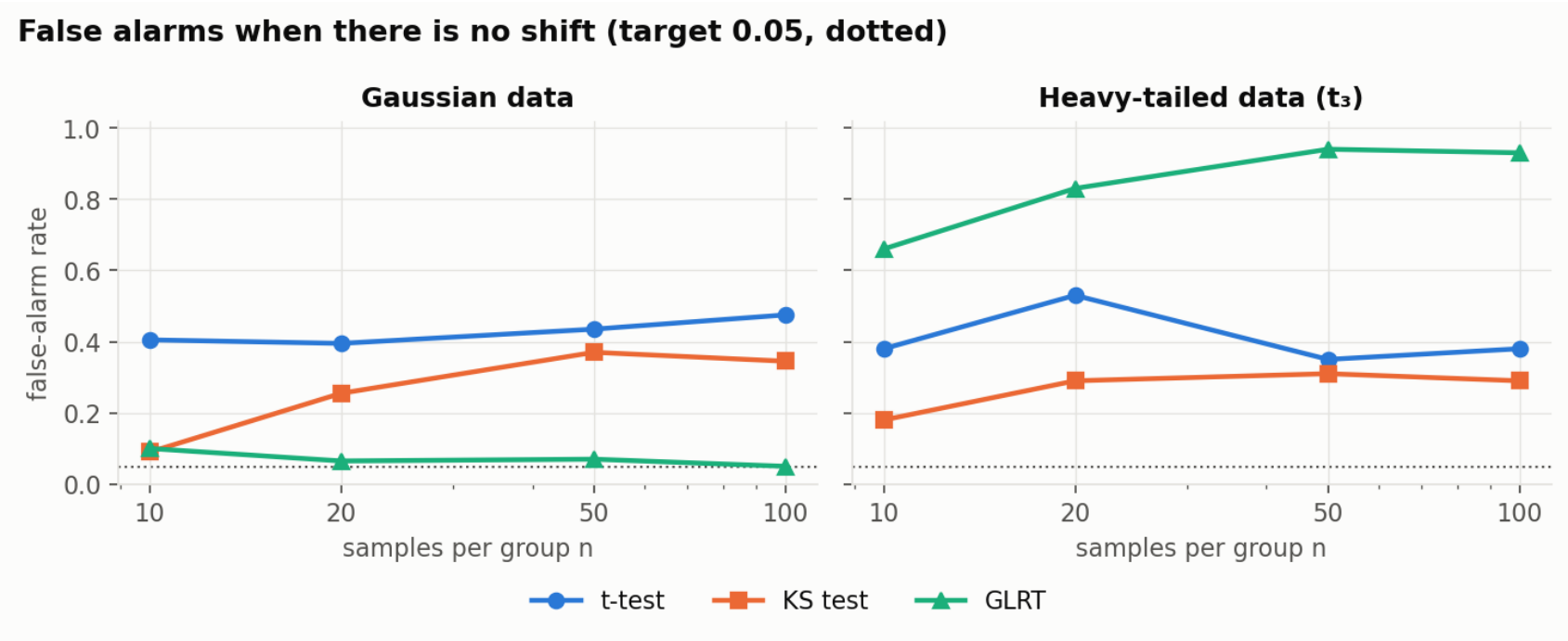
Reference distribution and dimensionality

Wilks' theorem: under H_0 and for large N, $G \approx \chi^2_r$, where r is the number of extra free parameters H_1 has over H_0.

Model	params per Gaussian	r	$d = 10$
Full covariance	$d + \frac{d(d+1)}{2}$	$d + \frac{d(d+1)}{2}$	65
Diagonal	$d + d$	$2d$	**20**

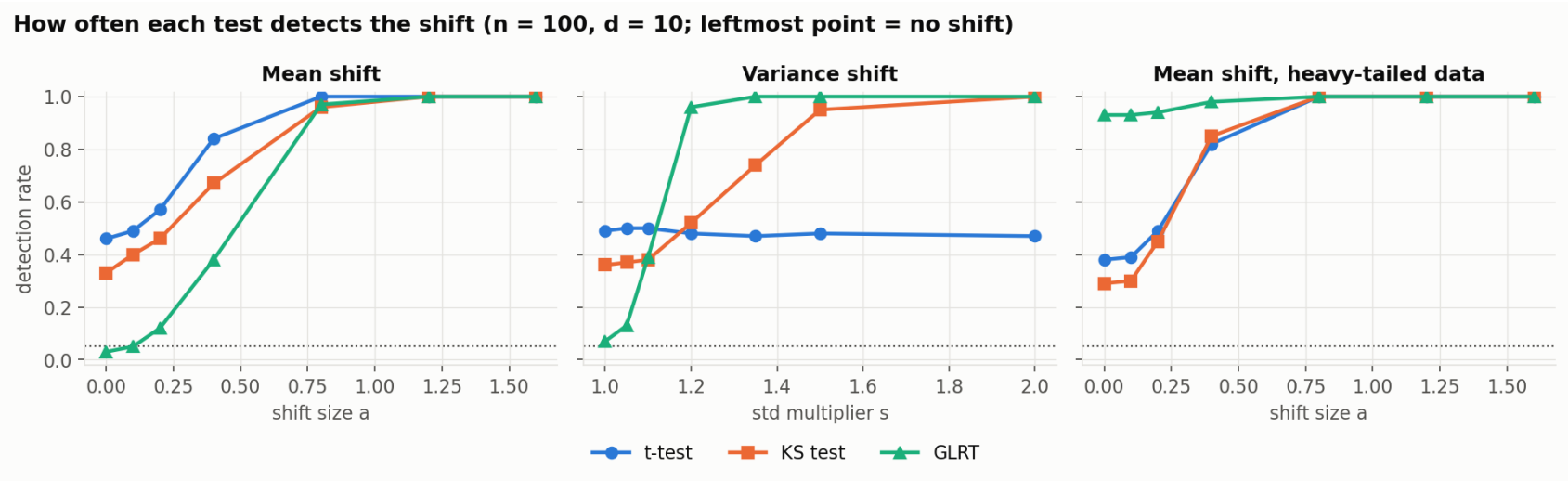
- The diagonal model is why our test uses χ^2_{20}.
- **Dimensionality cost:** the full model estimates $\sim d^2/2$ covariance entries per sample.
  - $\hat{\Sigma}_X$ has rank $\leq n - 1$, so $|\hat{\Sigma}_X| = 0$ and G is undefined unless $n > d$.
  - Wilks needs $N \gg r$. With $r = 65$ and $n = 20$ the χ^2 approximation is poor.
- Wilks also assumes the Gaussian model is **correct**. Heavy tails break it, which is why the heavy-tailed false alarms reach 93%.
- Fix: keep G but get its null distribution by **permutation** (shuffle X/Y labels, recompute G).

Results: False Alarms (no shift)



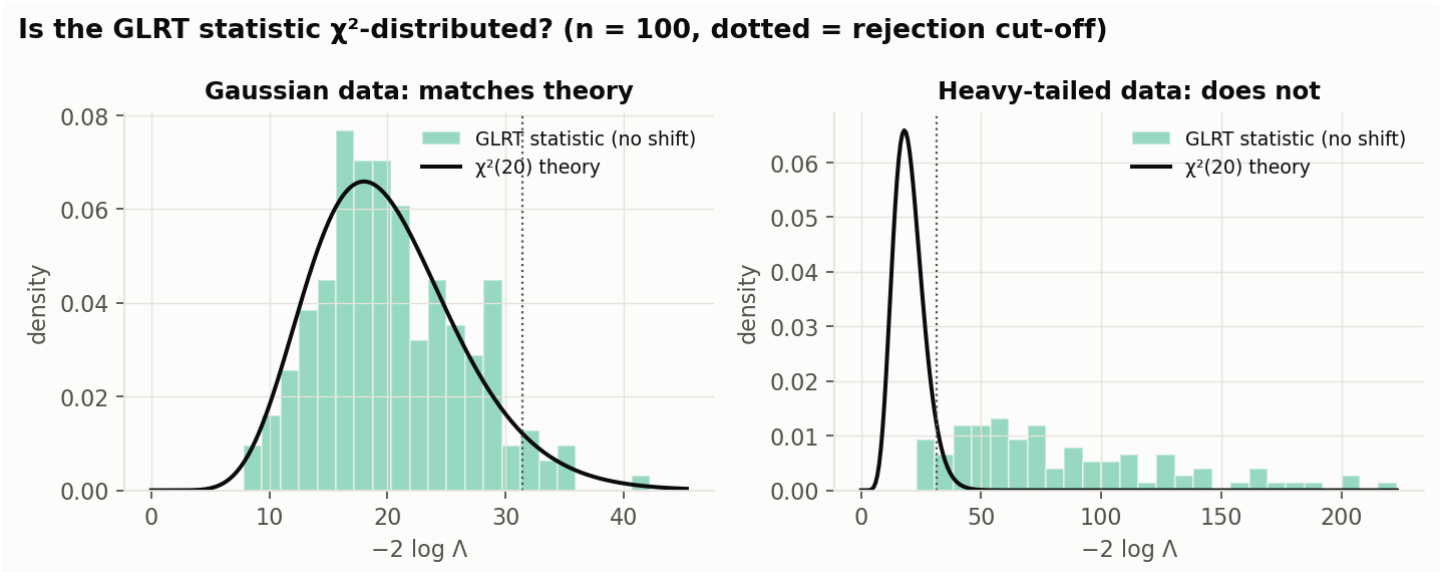
- t-test and KS: 30–50% false alarms, because 10 features at 5% each give $1 - 0.95^{10} \approx 40\%$.
- GLRT on Gaussian data: about 5%, as it should be.
- GLRT on heavy-tailed data: up to 93% false alarms.

Results: Detection Rate ($n = 100$)



- **Mean**: all three tests reach about 100% by $a = 0.8$.
- **Variance**: the t-test cannot see it; GLRT is best (96% vs 52% for KS at $s = 1.2$).
- **Heavy-tailed**: GLRT reports a shift even with no shift (93%), so its detections are meaningless.

Results: Does the GLRT Follow χ^2_{20}?



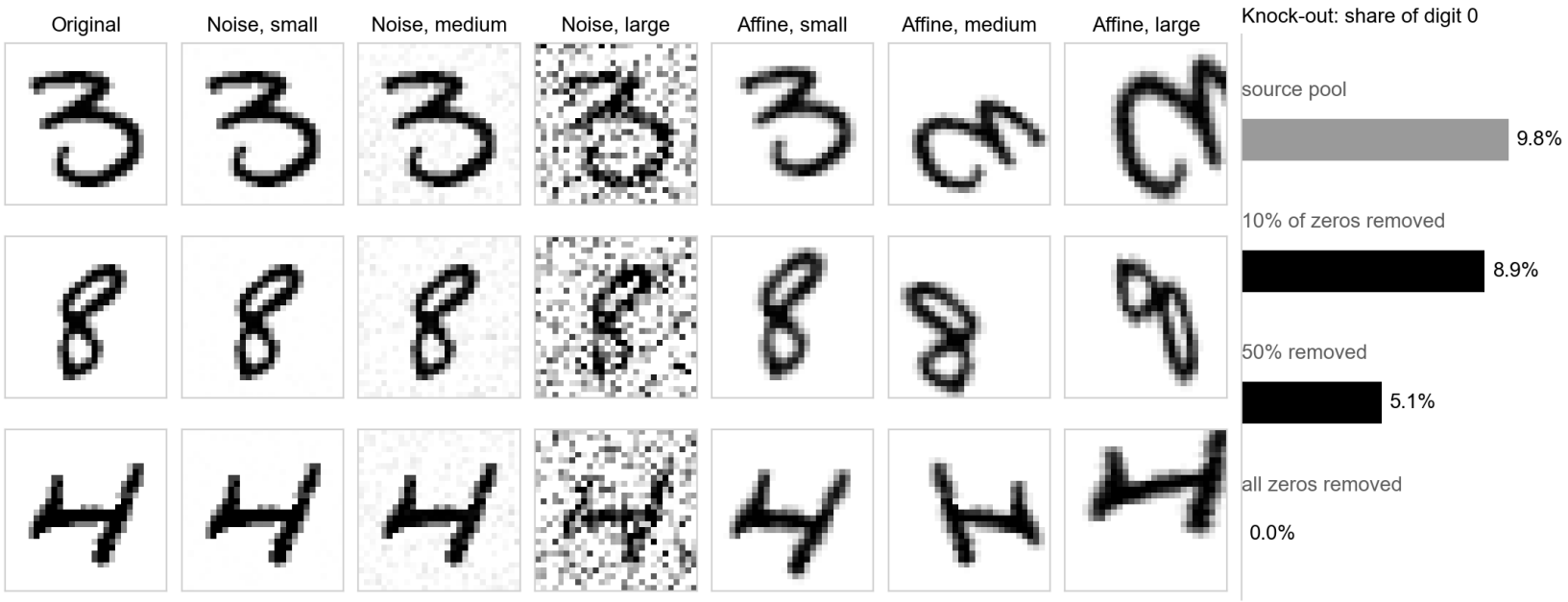
- Gaussian data: matches χ^2_{20}, so its p-values
- Heavy-tailed data: values are far above χ^2_{20}, so ordinary noise looks like a shift.

Results: Summary ($n = 100$)

	No shift (false alarm)	**Mean** $a = 0.8$	**Variance** $s = 1.2$	**Heavy-tailed (no shift)**
t-test	0.48	1.00	0.48	0.38
KS	0.35	0.96	0.52	0.29
GLRT	**0.05**	0.97	**0.96**	0.93

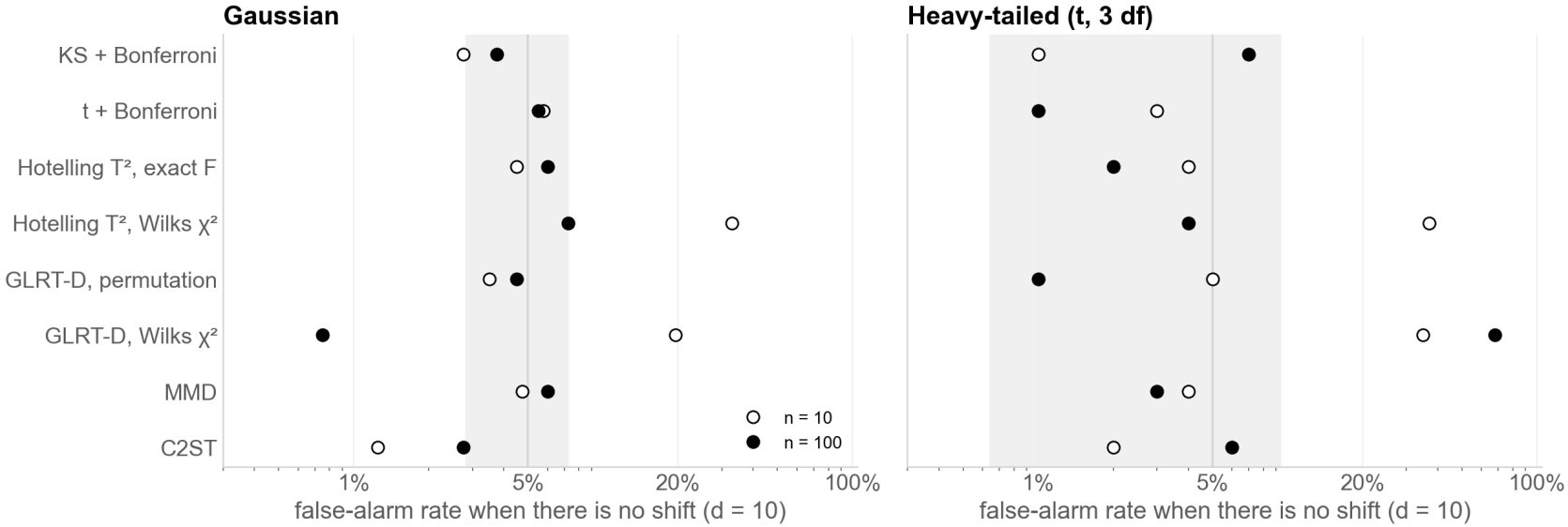
- GLRT is the best test when the data is Gaussian, but fails when the Gaussian assumption is violated.
- Testing each feature separately without a multiple-testing correction inflates the false alarm rate.
- The GLRT requires a suitable distributional model for its reference distribution to be valid.

Comparison: MNIST shifts (Failing Loudly)



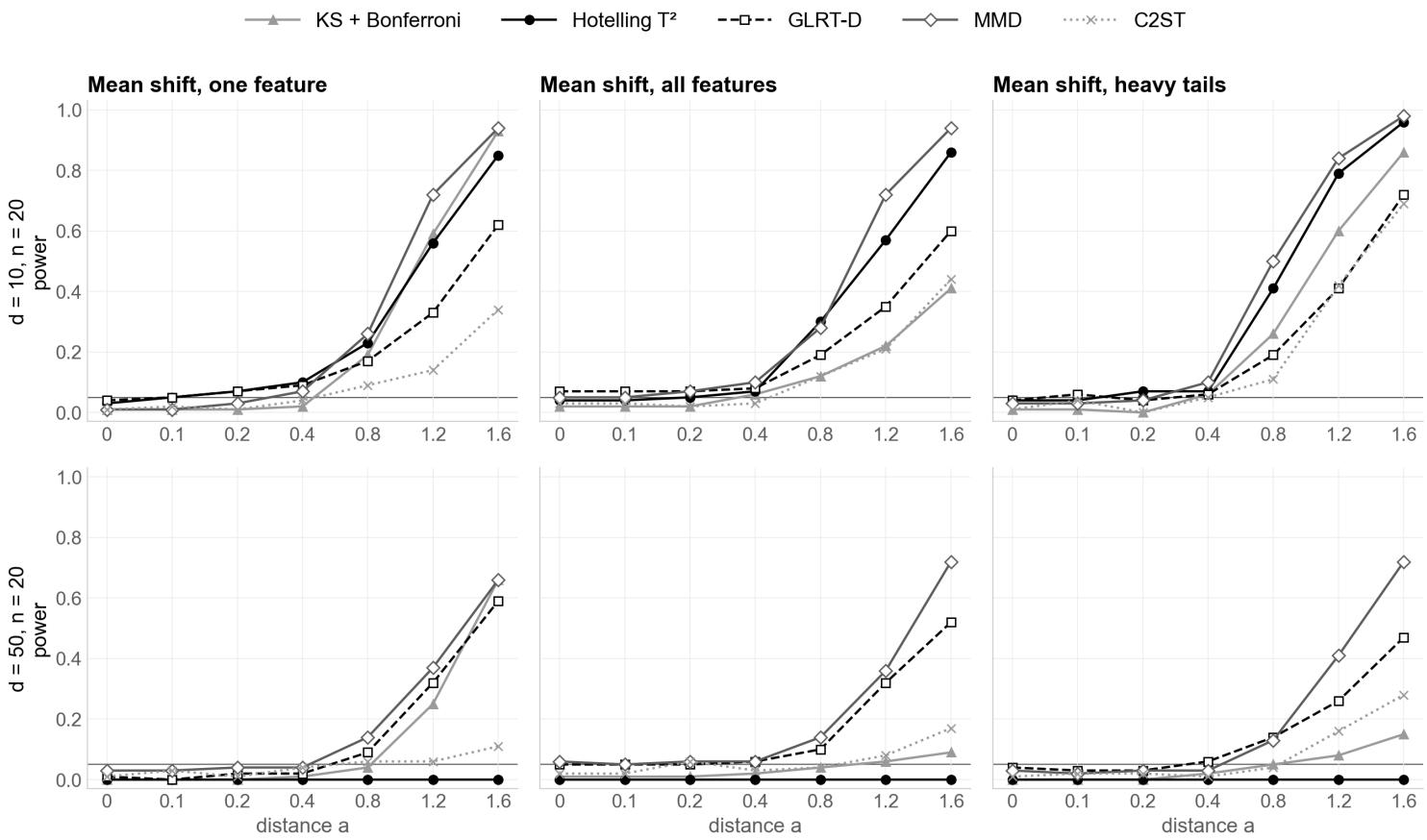
- **Noise**: Gaussian pixel noise, $\sigma = 1, 10, 100$ out of 255.
- **Affine**: random rotation, shift, shear, zoom (plus flips).
- **Knock-out**: remove some 0s; only the mix of digits changes.
- Each hits 10%, 50% or 100% of images. Tests see 784 pixels reduced to 32 (PCA) or 9 (BBSD).

Comparison: False alarms with corrections



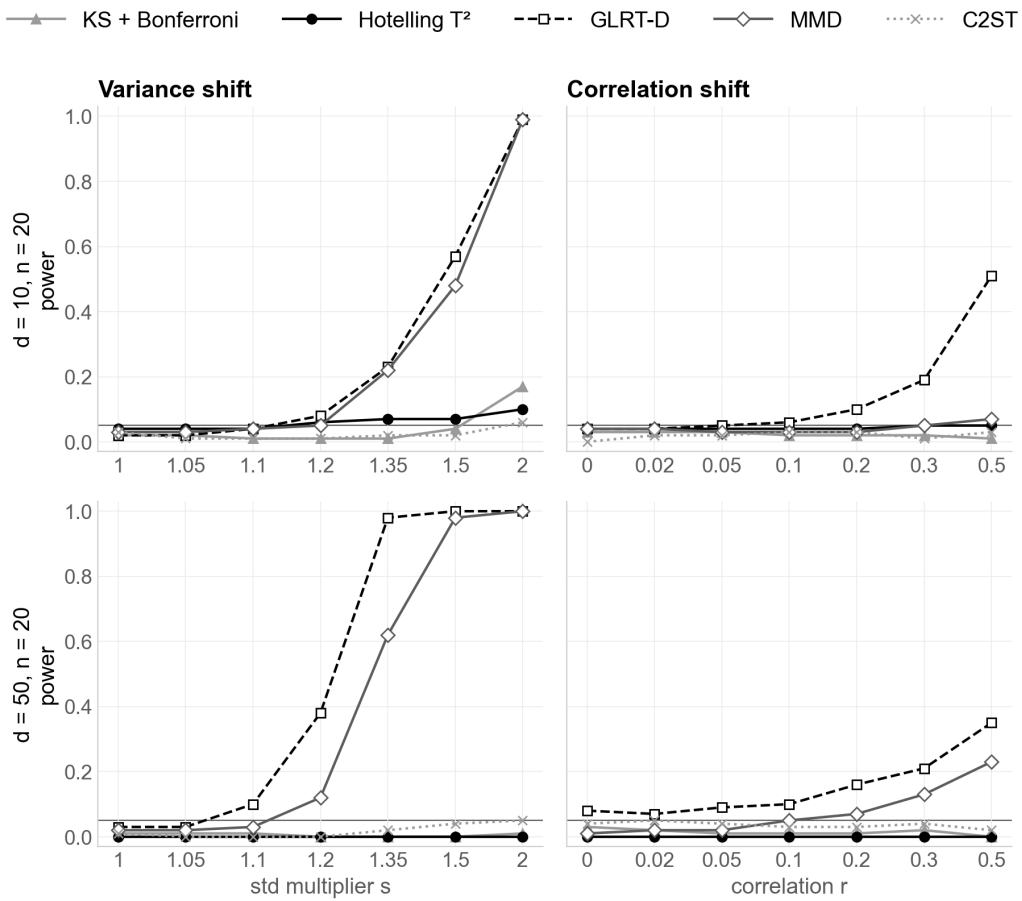
- With Bonferroni ($p_{\min} \times d$), t and KS hold 5% (vs 30–50% without it).
- Hotelling (exact F) and permutation GLRT-D hold 5%, even on heavy tails.
- Wilks' χ^2 fails: 33% (Hotelling) and 20% (GLRT-D) at $n = 10$; 68% on heavy tails at $n = 100$.

Comparison: Mean shifts ($n = 20$)



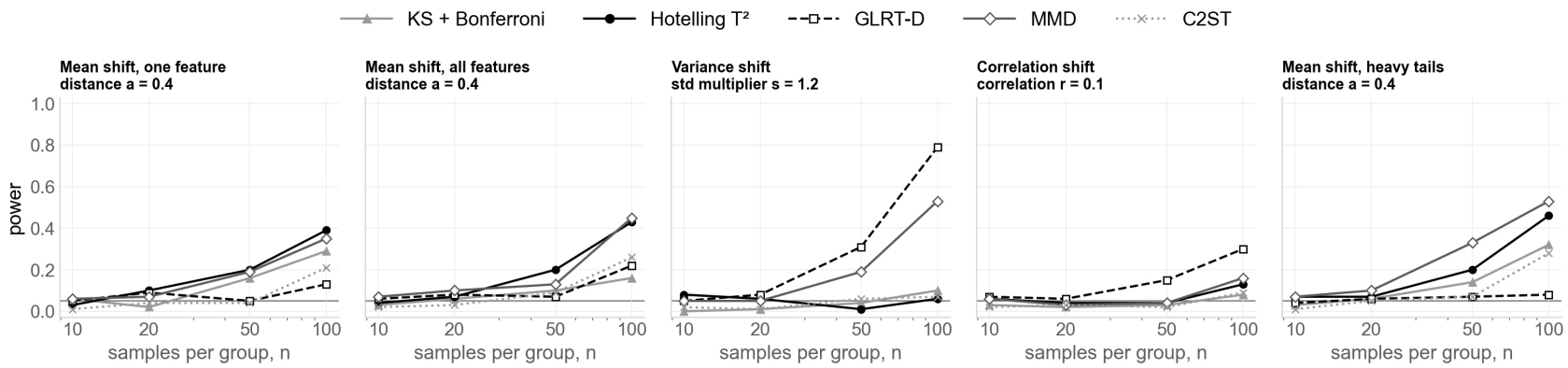
- One feature moved: KS + Bonferroni keeps up. Spread over all features: it falls behind.
- Hotelling pays for estimating Σ; at $d = 50$, $n = 20$ it is undefined ($2n - 2 < d$).

Comparison: Variance and correlation ($n = 20$)



- Mean tests (Hotelling) are blind: no mean moves.
- Correlation keeps every feature $\mathcal{N}(0, 1)$, so per-feature KS cannot see it. Only GLRT-D, which models Σ, detects it.

Comparison: Power against sample size ($d = 10$)



- More samples help only the tests that can see the change: on correlation, KS and Hotelling stay at 5% even at $n = 100$.
- At these moderate shifts, only GLRT-D on variance passes 50% power by $n = 100$ (0.79).
- Heavy tails stall GLRT-D (about 8%) while MMD climbs to 0.53.

Comparison: Which test, when

If you expect	**Use**	**Because**
Mean shift in a few features	t or KS + Bonferroni	All power on the moved feature
Mean shift over many features	Hotelling ($2n \gg d$) or MMD	Pools weak evidence
Variance, correlation, label shift	GLRT-D (GLRT-C for variance)	Models the covariance
Nothing specific / heavy tails	MMD	No model to break
Any GLRT	Permutation p-value	Wilks gave up to 94% false alarms
Under $\sim$100 samples	Not C2ST	Spends half the data on training